

BE 101P: Reverse Engineering for Bioengineers

Course Code : BE101P

Course Name : Reverse Engineering for Bioengineers

L-T-P-C: 0-0-2-1

Intended for : B.Tech.-M.Tech. Integrated Dual Degree in Bioengineering (Core Course)

Prerequisite : No prerequisite (compulsory for Bioengineering students)

Mutual Exclusion : NA

Course Modules :

The students focus on hardware reverse engineering (RE). In the process of RE students understand existing technologies, functions, features, objects, components and systems. By carefully disassembling, observing, testing, analyzing and reporting, students can understand how something works and suggest ways it might be improved. This process requires careful observation, disassembly, documentation, analysis and reporting. Many times, the reverse engineering process is non-destructive. This means that the object or component can be reassembled and still function just as it did before it was taken apart. Throughout the reverse engineering project, the students are able to think of ways these objects could be improved. Is there some way it could function better? or manufactured less expensively? The students will use observations to make suggestions for improvement of the product. (28 Hours)

Learning Topics:

Reverse Engineering of Biotechnological/Biomedical- Devices/ prototypes.

Textbooks:

1. Lam, R.H. and Chen, W., **Biomedical Devices: Materials, Design, and Manufacturing**, Springer, 2019.
2. Boccato C, Cerutti S, Vienken J, editors, **Medical devices: improving health care through a multidisciplinary approach** [Internet]. Springer International Publishing, 2022

References:

1. Sokovic and Kopac, **RE as necessary phase by rapid product development**, Journal of Materials Processing Technology, 2005
2. Eldad Eilam, **Reversing: Secrets of Reverse Engineering**, Wiley, 2005.
3. Chris Eagle, **The IDA Pro Book: The Unofficial Guide to the World's Most Popular Disassembler**.

Approval: 49th BoA